A 72-GS/s, 8-Bit DAC-Based Wireline Transmitter in 4-nm FinFET CMOS for 200+ Gb/s Serial Links

Timothy O. Dickson, *Senior Member, IEEE*, Zeynep Toprak Deniz, *Senior Member, IEEE*, Martin Cochet, *Senior Member, IEEE*, Troy J. Beukema, Marcel Kossel, *Senior Member, IEEE*, Thomas Morf, *Senior Member, IEEE*, Young-Ho Choi, Pier Andrea Francese, *Senior Member, IEEE*, Matthias Brändli, Christian W. Baks, Jonathan E. Proesel, *Senior Member, IEEE*, John F. Bulzacchelli, *Member, IEEE*, Michael P. Beakes, *Member, IEEE*, Byoung-Joo Yoo, *Member, IEEE*, Hyoungbae Ahn, *Member, IEEE*, Dong-Hyuk Lim, Gunil Kang, Sang-Hune Park, Mounir Meghelli, *Member, IEEE*, Hyo Gyuem Rhew, Daniel J. Friedman, *Fellow, IEEE*, Michael Choi, Mehmet Soyuer, *Life Fellow, IEEE*, and Jongshin Shin, *Member, IEEE*

***Abstract*— This article details the design and measurement of a digital-to-analog converter (DAC)-based source-series terminated (SST) transmitter (TX) for wireline applications in 4-nm FinFET CMOS technology. The DAC achieves 8-bit resolution and high analog output bandwidth by using a segmented architecture along with a single-ended LSB. Strength adjustment of the lower four DAC LSBs relative to the upper four DAC MSBs is accomplished with a hybrid analog/digital tuning approach, which overcomes minimum device-size limitations that can limit the effectiveness of pure digital tuning for SST drivers. The resulting DAC design achieves well-matched MSB/LSB segments with −0.63/0.67 LSB integral nonlinearity (INL) and −0.16/0.43 LSB differential nonlinearity (DNL). Time-domain modulation of 216-Gb/s PAM8 and frequency-domain modulation of 212-Gb/s orthogonal frequency-division multiplexing (OFDM) are reported, demonstrating the capability of CMOS DACs to support frequency-domain modulation for wireline applications. The TX consumes 288 mW from a 0.95-V power supply.**

***Index Terms*— CMOS, digital-to-analog converter (DAC), orthogonal frequency-division multiplexing (OFDM), pulse amplitude modulation (PAM), serializer–deserializer (SerDes), source-series terminated (SST), transmitter (TX), wireline.**

Manuscript received 26 August 2022; revised 3 November 2022; accepted 1 December 2022. Date of publication 23 December 2022; date of current version 28 March 2023. This article was approved by Associate Editor Borivoje Nikolić. *(Corresponding author: Timothy O. Dickson.)*

Timothy O. Dickson, Zeynep Toprak Deniz, Martin Cochet, Troy J. Beukema, Christian W. Baks, John F. Bulzacchelli, Mounir Meghelli, Daniel J. Friedman, and Mehmet Soyuer are with the IBM Thomas J. Watson Research Center, Yorktown Heights, NY 10598 USA (e-mail: todickso@us.ibm.com).

Marcel Kossel, Thomas Morf, Pier Andrea Francese, and Matthias Brändli are with IBM Research Europe, 8803 Rüschlikon, Switzerland.

Young-Ho Choi, Byoung-Joo Yoo, Hyoungbae Ahn, Dong-Hyuk Lim, Gunil Kang, Sang-Hune Park, Hyo Gyuem Rhew, Michael Choi, and Jongshin Shin are with Samsung Electronics, Hwaseong 18448, Republic of Korea.

Jonathan E. Proesel was with the IBM Thomas J. Watson Research Center, Yorktown Heights, NY 10598 USA. He is now with Nubis Communications, New Providence, NJ 07974 USA.

Michael P. Beakes, retired, was with the IBM Thomas J. Watson Research Center, Yorktown Heights, NY 10598 USA. He resides in Yorktown Heights, NY 10598 USA.

Color versions of one or more figures in this article are available at https://doi.org/10.1109/JSSC.2022.3228632.

Digital Object Identifier 10.1109/JSSC.2022.3228632

I. Introduction

SERIAL link data rates have steadily increased over the past two decades, fueled by the need for higher bandwidth network connectivity in data centers and servers. The trend is exemplified through an examination of the per-lane data rate evolution in Ethernet standards. Starting with IEEE 802.3ae 2.5-Gb/s lane rates in 2003 and leading to IEEE 802.3ck 100-Gb/s lane rates in 2022 [1], these rates have doubled approximately every four years. As this trend continues, interest in 200-Gb/s serial links is emerging. To support such high data rates, high-speed wireline transmitters (TXs) with equalization are required. Fig. 1 illustrates two popular high-level architectures for wireline TXs. In the analog architecture of Fig. 1(a), feed-forward equalization (FFE) is implemented within the high-speed analog portion of the TX. Unit-interval (UI) spaced FFE tap delays are typically achieved using custom digital delay elements, such as latches clocked with multiple phases of half- or quarter-rate clocks. FFE tap weights are programed either by current adjustments in a current-mode logic (CML) driver, as shown in Fig. 2(a) (e.g., [2]), or by assignable segments in a voltage-mode [source-series terminated (SST)] driver, as shown in Fig. 2(b) (e.g., [3]). Both approaches can support four-level pulse amplitude modulation (PAM4) by adding parallel driver segments to transmit MSB and LSB data symbols. As the industry has moved to PAM4 signaling, the power and complexity overhead of incorporating additional driver segments and FFE delay generation circuitry in the TX analog domain has resulted in a shift to the digital-to-analog converter (DAC)-based TX architecture shown in Fig. 1(b). In applications requiring a high FFE tap count (perhaps five or more), DSP implementation of PAM4 modulation with FFE can simplify the high-speed analog data path and avoid excessive capacitive loading at the TX output that degrades analog output bandwidth.

For 200+ Gb/s serial links, DAC-based wireline TX architectures offer flexibility to support more advanced modulation in the time domain, such as six or eight level PAM, as well as

0018-9200 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

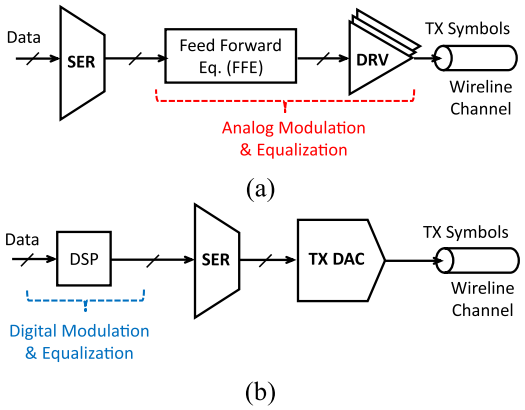


Fig. 1. Illustrative wireline TX architectures: (a) analog TX and (b) DAC-based TX.

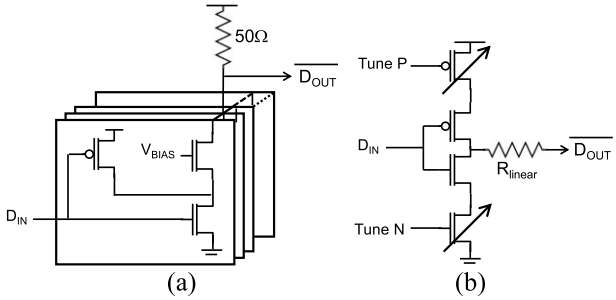


Fig. 2. Wireline TX driver topologies: (a) tailless CML [2] and (b) SST [3].

in the frequency domain with orthogonal frequency-division multiplexing (OFDM). By using higher density constellations, such as QAM64, OFDM communication systems require lower DAC and analog-to-digital converter (ADC) sampling rates to achieve the same line rate compared to PAM4. However, CMOS DACs with sufficient linearity to support OFDM have to date not been demonstrated for electrical serial links.

The output driver topology is another key consideration in wireline TX design. A literature review of state-of-the-art high-speed TXs shows numerous CML-based designs achieving symbol rates between 64 and 112 GBaud in CMOS technology nodes ranging from 40 down to 10 nm [2], [4], [5], [6]. In comparison, SST-based designs have, thus far, achieved a maximum speed of 56 GBaud in a 7-nm technology [7]. Compared to CML topologies, the SST topology of Fig. 2(b) is more amenable to integration in highly digital serializer–deserializer (SerDes) due to its simpler biasing. Moreover, it provides superior electrostatic discharge (ESD) protection to charged-device model (CDM) events since the linearization resistor R_{linear} attenuates the ESD pulse before it reaches the driver active devices. Increasing the speed of SST-based wireline TXs is, therefore, desirable.

This article reports the design and measurement of a 72-GS/s 8-b DAC-based wireline TX in 4-nm FinFET CMOS to support higher order digital modulation formats for wireline applications [8]. Techniques for realizing 8-b DAC resolution with a low number of DAC elements (SST driver segments) are presented, which is key for achieving high TX analog output bandwidth. System-level design and modulation

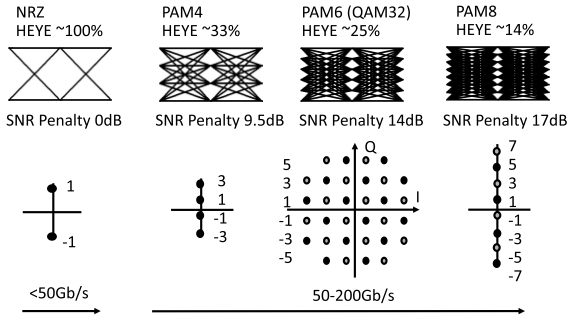


Fig. 3. Symbol constellations for wireline time-domain modulation.

considerations for 200+ Gb/s signaling are discussed in Section II. Section III describes the TX architecture and building blocks. Experimental results, including measurements for both time- and frequency-domain modulations, are presented in Section IV. Finally, Section V draws conclusions.

II. DAC System Design and Modulation

The DAC system design targets support of 200+ Gb/s data rates with applications ranging from short-reach links, such as chip-to-optical-module channels, to long-reach links, such as passive copper Twinax cables for intra-rack connectivity. Such a high data rate requires consideration of specific modulation and coding approaches together with the practical limitations of both the channel characteristics (such as dispersion and crosstalk) and silicon technology capabilities. Primary concerns related to advanced CMOS technology nodes involve achieving high frequency and low-jitter clocking, and supporting sufficient linear dynamic range (i.e., maximum achievable signal swings versus noise). For data modulation, both time-domain PAM and frequency-domain OFDM with QAM subcarriers were evaluated. The primary rationale for considering OFDM modulation is the potential for subchannel-carrier-based modulation to better support high-density constellations, such as QAM32 or QAM64 (equivalent to six- or eight-level PAM in the time domain, as seen in Fig. 3) for improved spectral efficiency in the presence of time jitter and residual intersymbol interference (ISI) arising from unequalized channel time dispersion. The improved spectral efficiency of high-density constellations lowers the required bandwidth for a target data rate. For the DAC design, a baud rate in the range of 80 GBaud and associated bandwidth of 40 GHz was originally targeted, requiring 2.5-b/s/Hz coded spectral efficiency for a 200-Gb/s payload. This level of spectral efficiency is only capable of being supported with six- or eight-level PAM, QAM32, or QAM64. The 8-b resolution of the DAC was chosen based on coded link capacity SNR of 2.5 b/s/Hz (resulting in ∼20-dB SNR at sensitivity assuming a 5-dB loss from capacity code) with an assumption of ∼12-dB dynamic range overhead for OFDM peaks and an additional ∼12-dB budget for OFDM subcarrier power loading.

Subcarrier-based modulation has the potential to better handle high-density constellations compared to conventional time-domain PAM modulation due to two primary reasons: edge effects and residual ISI. Referring to Fig. 3, as the spectral

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

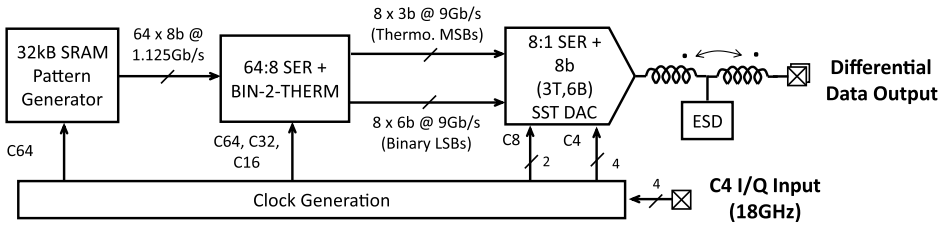


Fig. 4. DAC-based TX test chip architecture.

density of PAM is increased, the horizontal eye margin shrinks rapidly due to edge crossover effects arising from channel bandwidth limitations, resulting in a very narrow timing window over which the signal can be recovered and increasing the sensitivity of the system to degradation from clock jitter, which maps the edge transitions to amplitude noise through PM-to-AM conversion. As the constellation density increases, the system also becomes more susceptible to degradation from residual ISI. Residual ISI peaks are effectively amplified by the modulation SNR penalty, which is 9.5 dB for PAM4 and grows to 17 dB for PAM8. Therefore, a PAM-based system using a high-density constellation will require extremely precise low-jitter clocking and accurate equalization, and can be significantly degraded by channel dispersion outside the system's equalization range. An N-subcarrier OFDM design solves these problems by lowering the QAM constellation symbol rate by a factor of N, where N may range from 32 to 256 in a practical system. This reduction decreases the channel time dispersion relative to the symbol period by the same factor of N, effectively eliminating degradation from edge effects and residual ISI when an appropriate cyclic prefix is added to the OFDM frames to maintain subchannel orthogonality in the presence of ISI [9]. The OFDM system has a further advantage as it is possible to independently power load the OFDM subcarriers to equalize SNR across frequency without impacting QAM subcarrier symbol equalization. This capability is not present in a time-domain PAM system, where frequency shaping and required channel equalization are not independent.

These benefits must be weighed against some practical concerns of using OFDM in wireline links. First, OFDM lacks backward compatibility with existing time-domain wireline standards. Second, the DSP required for FFT and inverse FFT (IFFT) functions will add power and latency to the serial link. Third, OFDM requires additional cyclic prefix and frame pilot symbols to provide subchannel amplitude and phase references. Finally, the peak-to-average ratio of OFDM signaling can range from 6 to 10 dB. Consequently, DACs and ADCs in electrical transceivers may require 1- or 2-bit higher resolution than those for time-domain transceivers [9]. While addressing the first three concerns is beyond the scope of this work, this article investigates DAC-based TX design considerations to achieve sufficient bandwidth and linearity in advanced FinFET CMOS process nodes to support OFDM applications.

III. Transmitter Design

The block diagram of the DAC-based TX test chip is depicted in Fig. 4. The design employs an 8-b DAC; link analysis indicates this resolution is sufficient for the reach targets described in Section II when used in conjunction with an 8-b ADC [10]. Modulated, pre-equalized 8-b symbols are loaded into a 32-kB SRAM. A pattern generator sequencer clocked at a C64 rate (1.125 GHz or 1/64th of the TX output symbol rate) feeds data from the SRAM into the TX 64:8 serializer. Within this serializer, the upper two MSBs of the binary-weighted data are thermometer encoded. The segmented 1/8th rate data are then applied to the input of the DAC macro, which includes the final 8:1 serialization and SST driver. The test chip receives quadrature quarter-rate 18-GHz clocks, which are internally divided to generate all required sub-rate clocks for transmit serialization. This section describes the design of the high-speed clock and data path circuitry in the TX.

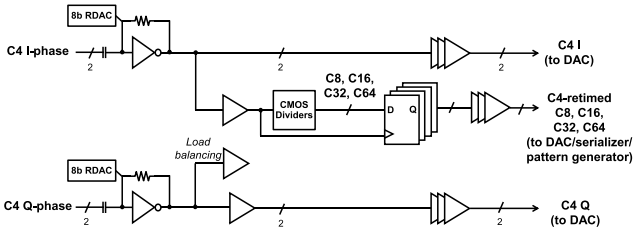


Fig. 5. TX test chip clock generation and buffering.

A. Clock Generation

In the clock generation block of Fig. 5, external quadrature quarter-rate clocks are applied to ac-coupled CMOS inverters. The dc level of the clock signal at the input to the inverters is adjusted via 8-b R-2R voltage DACs in order to correct for duty cycle distortion on both the I and Q phases in a manner similar to that described by Menolfi et al. [11]. CMOS frequency dividers produce all required sub-rate clocks for serialization and pattern generation by dividing the C4 I phase. As seen in Fig. 5, dummy loading is placed on the C4 Q path to prevent incurring systematic I/Q phase error. The sub-rate clocks, from the eighth rate (C8, or 9 GHz) down to the 64th rate (C64, or 1.125 GHz), are retimed to the rising edge of the C4 I clock phase to maximize serializer timing margins.

B. DAC and High-Speed Data Path

To maximize analog output bandwidth in a wireline SST-based DAC, the number of SST driver segments must be minimized. In this work, the objective is to achieve 8-b DAC resolution with far fewer elements than in a conventional 255 unit-element DAC to lower the capacitive self-loading

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

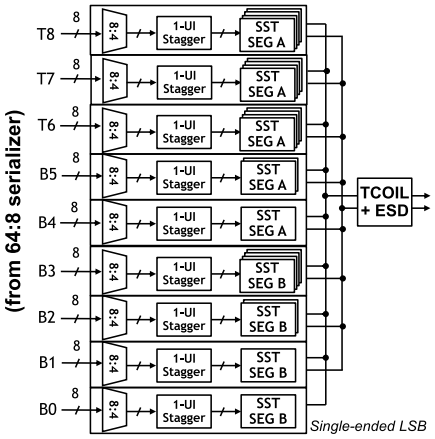


Fig. 6. Segmented DAC architecture.

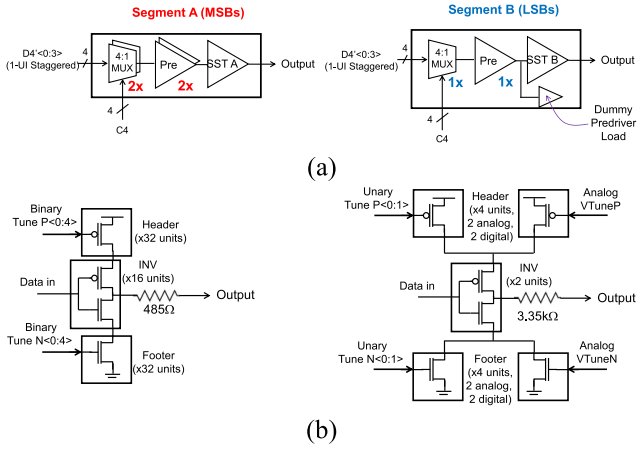


Fig. 7. (a) DAC segments A and B. (b) Corresponding SST driver segments.

at the DAC output. The architecture of this 8-b segmented DAC is shown in Fig. 6. Eighth-rate single-ended data from the 64:8 serializer is fed into 8:4 MUX blocks within the DAC. The resulting differential quarter-rate data are then staggered by one UI before reaching the DAC segments. The design incorporates two different SST output driver segments, referred to as segments A and B. The former is used in the upper four DAC MSBs, while the latter is associated with the lower four DAC LSBs. Segment A has a nominal drive strength that is eight times larger than that of segment B. In addition, the DAC LSB segment employs a single-ended segment B to add one extra bit of DAC resolution without doubling the segment count and ratio of segment B to segment A impedance. Similar approaches have previously been found in CDACs for SAR ADCs [12]. For a wireline TX DAC, this architecture choice comes at the expense of slightly higher common-mode output variation (approximately half of an LSB voltage magnitude or about 1.5–2 mV). In total, 256 DAC analog output levels are realized with only 15 A segments and eight B segments.

As seen in Fig. 7(a), each DAC segment consists of the final 4:1 MUX, a predriver, and a corresponding SST driver segment whose single-ended schematic is shown in Fig. 7(b). This diagram also illustrates another important benefit of the A/B driver segment approach that the low segment count reduces the number of required 4:1 MUX blocks and lowers the C4 clocking power. The A segments use a MUX and predriver that has twice the drive strength of the corresponding blocks in the B segments. Since the SST driver segment strengths vary by an 8:1 ratio, a dummy load is added to the output of the predriver in the B segments to maintain a 2:1 load ratio at the output of the A and B predrivers. These techniques manage loading on the data path without building up systematic skew between the DAC MSB and LSB.

Tuning of the SST driver segments in Fig. 7(b) is necessary to realize not only a 50-Ω driver output impedance across process corners but also the 8:1 strength ratio between A and B segments. In the design, active device sizes for data-switching FETs are scaled by a factor of 8 between MSB and LSBs as dictated by the relative matching requirements of these segments. However, minimum device sizes limit such scaling when applied to the digital header and footer devices used for impedance tuning. The coarse resistance of these minimum-sized digital switches hinders tuning capabilities and prevents proper binary scaling of the segment impedances as required for DAC applications. CML-based DACs are more amenable to analog tuning, as adjustments to driver bias currents can be made without altering the driver impedance. In SST-based DACs, pure analog tuning of driver strength through header and footer current sources can degrade TX linearity due to the finite output impedance of those current sources. Digital switching of header and footer devices results in larger gate overdrive voltage. Consequently, the header and footer transistors will remain in the linear region of operation over a wider range of output voltages compared to analog tuning. In this work, a hybrid digital and analog tuning approach is adopted. Digital tuning of the header and footer devices is applied in the A segments associated with DAC MSBs. The strength of the B segments is adjusted via a coarse digital control along with a fine analog control from a current source. Fig. 8 illustrates the segment linearity benefits of the proposed tuning approach by comparing simulated resistance behavior for different design options. As seen in Fig. 8(a), variation of the Segment A pull-up and pull-down resistances across DAC output voltages is dramatically reduced by tuning the header and footer device strengths via digital switches instead of analog current sources. Full digital tuning of the segment B impedance is impractical due to the limited number of header and footer devices. As seen in Fig. 7(b), the number of driver header and footer transistors is exactly scaled to achieve the required 8:1 resistance ratio between segments A and B, leaving only four gate fingers in the headers and footers. This reduces the digital control resolution to the point where the required A-to-B resistance cannot be maintained across process variations. Rather than increasing header and footer device counts and incurring additional parasitics, the design adopts the proposed hybrid analog/digital tuning in the B segments associated with the DAC LSBs. As seen in Fig. 8(b), this method yields modest improvement in segment B resistance variation compared to pure analog tuning. This variation has 16x less impact on the overall DAC linearity compared to the segment A resistance

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

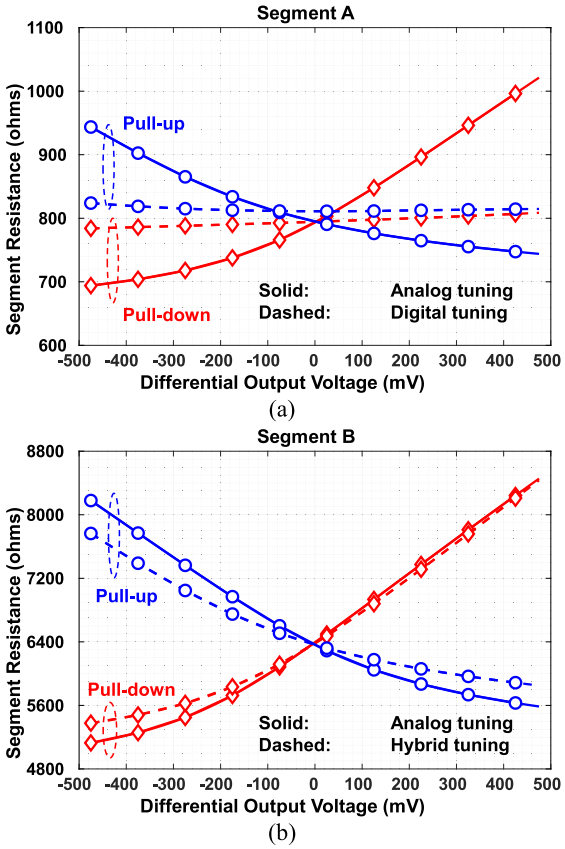


Fig. 8. Simulated resistances of DAC segments for different impedance tuning approaches. (a) Segment A analog and digital tuning. (b) Segment B analog and hybrid analog/digital tuning.

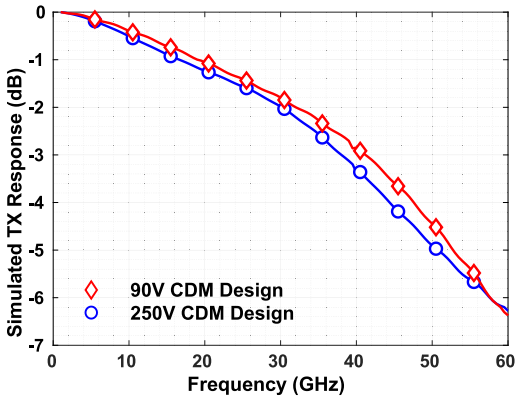


Fig. 9. Post-layout *RC* extraction simulation of the TX output frequency response including ESD diodes and T-coils for two different CDM protection levels.

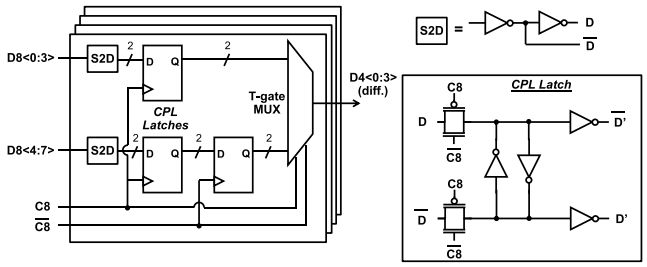


Fig. 10. 8:4 MUX schematic.

variation as the B segments are only used for the lower four DAC LSBs. Since the upper four DAC MSBs constitute approximately 94% of the driver conductance, these segments can be tuned based on the target driver output impedance. LSB segments are then tuned to achieve relative matching between the A and B segments. This hybrid analog and digital tuning approach is the key to reaching 8-bit DAC resolution with only 15 A segments and 8 B segments. DAC segment mismatch was found in the simulation to be dominated by global variation in the SST linearization resistors. These metal resistors account for ∼60% and ∼52% of the segment A and B resistances, respectively. Hence, a single calibration code is applied to all A segments, while a single calibration code and tuning voltages are shared amongst the B segments.

TX output matching networks, including ESD protection diodes and corresponding bandwidth-extension T-coils [13], were designed for CDM ESD protection levels of 250 and 90 V. The TX frequency response of the two designs, as reported in Fig. 9, was determined via step response simulations including an *RC*-extracted netlist of the DAC layout. The 250- and 90-V CDM protection level designs achieve 3-dB bandwidths of approximately 38 and 41 GHz, respectively. While both designs were fabricated, measurements of Section IV report results for the higher bandwidth design with a lower CDM protection level. Nonetheless, the simulated data of Fig. 9 demonstrate the capability of the SST-based DAC architecture to achieve both high analog output bandwidth and high CDM protection levels.

Turning to the remainder of the high-speed data path, Fig. 10 shows the schematic of the 8:4 MUX. It converts single-ended eighth-rate data from the 64:8 serializer to differential data, which is then applied to complementary pass transistor logic (CPL) latches shown in the bottom right of the figure. A transmission-gate MUX uses alternating edges of the C8 clock to produce the output quarter-rate data bits D4<0:3>, which are fed to the one-UI staggering circuit of Fig. 11(a) [2]. The circuit uses a network of CPL latches clocked at the C4 rate using quadrature phases to produce four quarter-rate data outputs D4'<0:3> that are progressively delayed by one unit interval. Doing so improves timing margins within the final 4:1 MUX, as will be discussed shortly. As seen in the timing diagram of Fig. 11(b), output data D4'<0>, D4'<1>, D4'<2>, and D4'<3> are aligned to the rising edges of the 0°, 90°, 180°, and 270° C4 clock phases, respectively.

Final 4:1 multiplexing is among the most challenging aspects of high-speed wireline TX design. The 4:1 MUX schematic shown in Fig. 12(a) adopts a transmission gate style MUX. Each of the four transmission gates is enabled for one UI to pass quarter-rate data through to the predriver input. Dual passgate logic (DPL) OR and AND gates derive one-UI clock pulses from the quadrature C4 clocks [7]. As shown in the timing diagram of Fig. 12(b), the clocking is arranged such that each of the one-UI staggered quarter-rate data is selected during the third out of four UIs for which the data

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

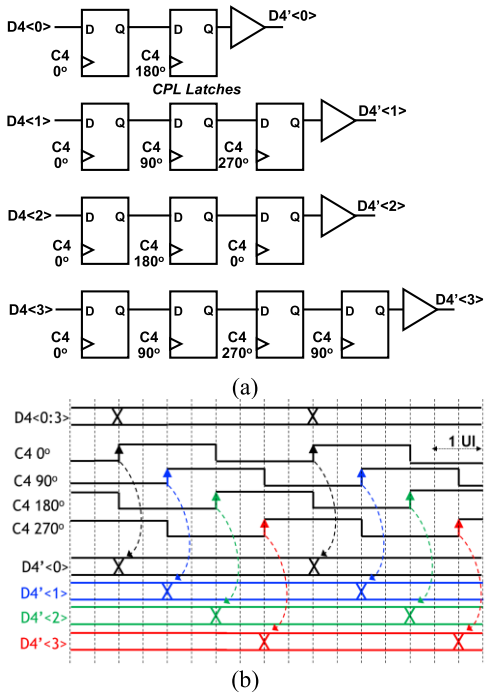


Fig. 11. (a) One-UI staggering circuit. (b) Timing diagram.

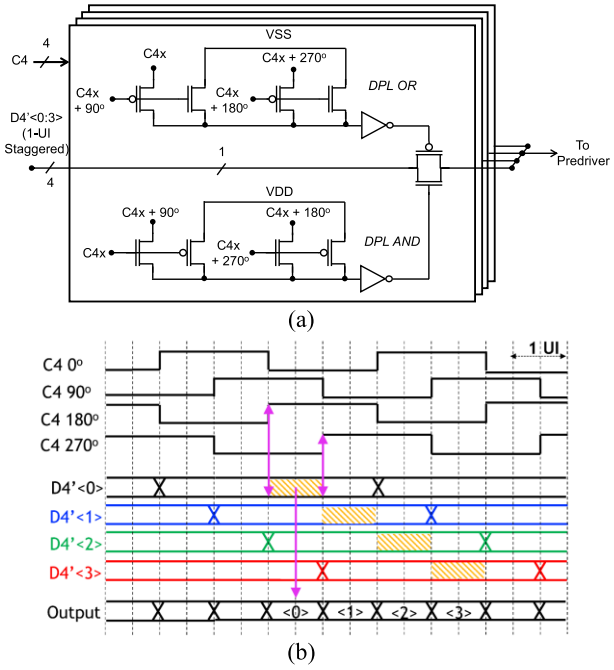


Fig. 12. (a) 4:1 MUX schematic. (b) Timing diagram.

are valid to ensure robust timing margins. Full-rate data are then fed into CMOS inverter-based predrivers, which produce rail-to-rail output data to the SST drivers. C4 clock duty cycle and quadrature accuracies are sensed at the output of a replica B driver segment. The input eighth-rate data pattern to this replica can be programmed as "10101010," "11001100," or "01100110" to sense clock I/Q errors or duty cycle errors in the I and Q clocks, respectively. In this test chip, the output of the replica is low-pass filtered and monitored off-chip to enable closed-loop calibration. Referring to Fig. 5, 8-b RDACs actuate duty cycle correction of the C4 clocks. As the test chip receives external quadrature clocks, I/Q separation is adjusted via an external mechanical phase shifter, as will be described in Section IV.

C. Serializer and Dynamic Segment Assignment

The 64:8 serializer block from Fig. 4 receives binary-weighted 8-b data from the pattern generator and produces segmented data, where the upper two DAC MSBs are encoded to three thermometer (unary) bits. Examination of a simple 2-b binary-to-3-b thermometer truth table reveals that two of the three thermometer encoded bits will be dc-imbalanced if balanced (or sufficiently random) binary data are applied. The lowest of the three thermometer encoded bits is a logic "1" for 75% of the 2-b binary input words, while the highest is a logic "1" for only 25% of the input words. This disparity has implications for the design of segmented DACs for wireline applications, as the DAC driver segments driven by dc-imbalanced thermometer encoded data will experience different switching activity factors and conduct different average currents. In turn, the DAC segments will experience thermal mismatch due to self-heating and will age at different rates, degrading DAC linearity and performance over time. In nanoscale CMOS technology nodes, driver active and passive device sizes, and interconnect widths must be increased to handle the worst case electromigration currents. Consequently, DAC driver parasitic capacitances will increase and degrade the TX output analog bandwidth. The situation is even more severe in OFDM applications, as OFDM's high peak-to-average ratio causes DAC segments driven by the upper and lower thermometer-encoded MSBs to switch for only a few percent of the transmitted symbols [9].

To prevent such a disparity in DAC MSB switching activity, a shuffling circuit embedded between the 32:16 MUX and the 16:8 MUX of Fig. 13 dynamically assigns thermometer-encoded data to the thermometer-encoded DAC MSBs. The shuffling is pseudo-randomly controlled through a 3-b PRBS9 circuit updated at a C64 rate (1.125 GHz), ensuring that data are shuffled at time intervals that are faster than self-heating time constants. As the PRBS9 state is updated and thermometer-encoded data are reassigned to different DAC MSBs, shuffling activity must occur without glitching data. Fig. 14 highlights the key circuitry of a 2:1 MUX processing thermometer encoded data within the 16:8 serializer. Thermometer encoded 16th-rate data T8_16', T7_16', and T6_16' are latched via L1 and L2 on the C16 clock rising edge, as is the STATE variable from the PRBS control circuitry via latch LS. Throughout the 16:8 MUX, latched 16th-rate data T8_16', T7_16', and T6_16' are buffered and distributed to selectors S1 and S2 in corresponding sub-2:1 MUXs. Control of the selectors is governed by the latched STATE variable, thus performing dynamic reassignment of the thermometer-encoded data to outputs of the 16:8 MUX. The selected data arrive at the input to latches L3 and L4 within their setup and hold times, ensuring glitch-free data switching. Unlike dynamic element matching [14], pseudo-random shuffling is

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

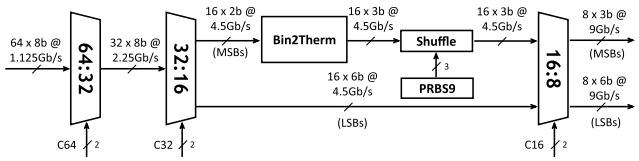


Fig. 13. 64:8 serializer with thermometer-encoded data shuffling.

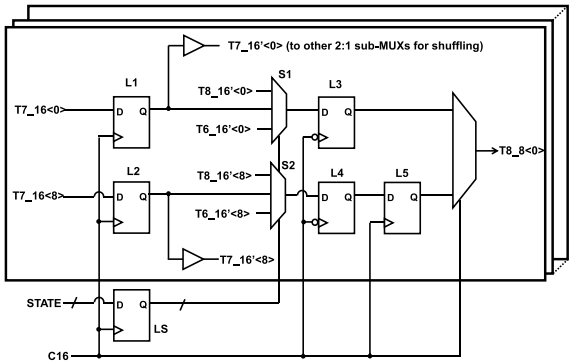


Fig. 14. Shuffling circuitry and 2:1 MUX processing thermometer encoded data within 16:8 MUX.

not intended to improve DAC linearity. Instead, the shuffling activity results in more data uniformity and lower worst case dc currents across all thermometer-encoded DAC segments, permitting the use of smaller transistor and resistor widths within the SST driver segments of Fig. 7(b) and reducing TX output capacitance while meeting electromigration requirements.

IV. Fabrication and Experimental Results

A. Test Chip and Experimental Setup

The TX test chip was fabricated in Samsung 4lpp 4-nm FinFET CMOS technology. Fig. 15(a) shows the die photograph, along with test chip layout details. The C4 pad-limited test chip occupies 1328 μm $\times$ 1420 μm, while the TX core along with ESD protection diodes and bandwidth extending T-coils measures 215.7 μm $\times$ 219.8 μm. Further layout details of the TX core are seen in Fig. 15(b). External quarter-rate I- and Q-phase clocks are fed into the centrally located clock receiver from the west and east edges of the TX core, respectively. Parallel data from the pattern generator flow from the 64:8 serializer on the north edge down to the DAC and TX output pads on the south edge.

The 4-nm test chip was flip-chip bonded to a GL102F 4-2-4 substrate, which was subsequently mounted on a Nelco4000-13 test card. TX outputs are taken from connectors at the top of the substrate through high-speed cables for direct examination on a Tektronix DSA8300 with 80E09B remote sampling modules. The 18-GHz clock input is derived from an Anritsu MG2697C frequency synthesizer, which sends the clock through a network of Krytar 4100400 180° hybrid couplers along with a Spectrum OPS-0002 mechanical phase shifter. The step response of the TX and test setup were measured, and the impulse response shown in Fig. 16(a) was

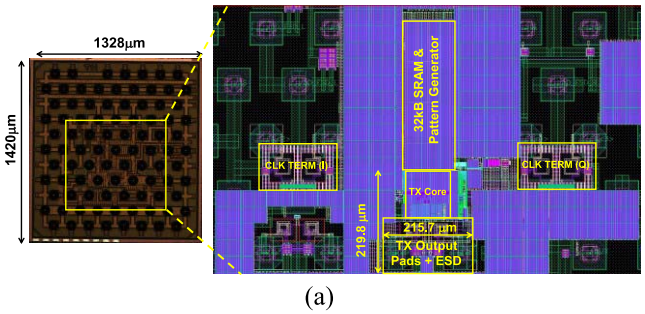


Fig. 15. (a) 4-nm test chip die photograph and layout details. (b) Layout details of the TX core.

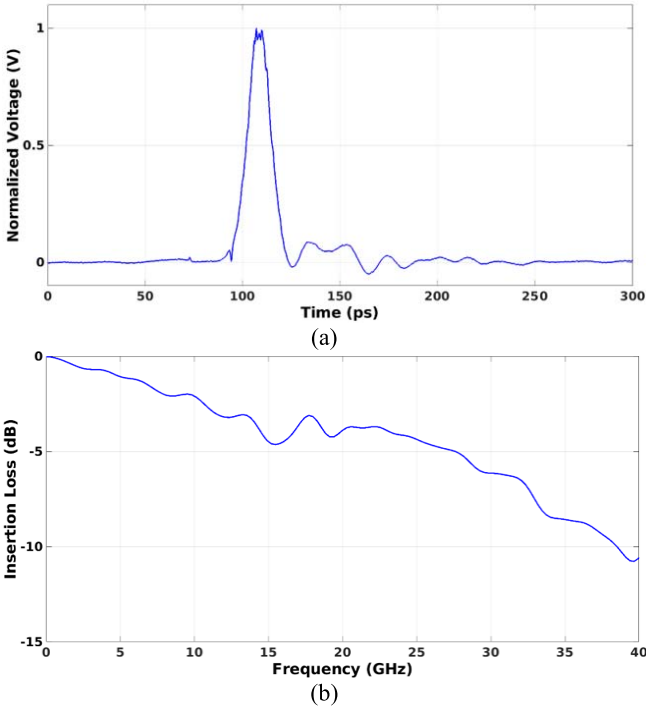


Fig. 16. (a) Measured impulse response of the TX and experimental setup. (b) Frequency response derived from the impulse response.

calculated from the derivative of the step response. Fig. 16(b) shows the computed frequency response, which shows 8.8 dB of loss at 36 GHz. This loss includes a measured 6-dB insertion loss from the cables, ac coupling capacitors, and oscilloscope remote sampling modules.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

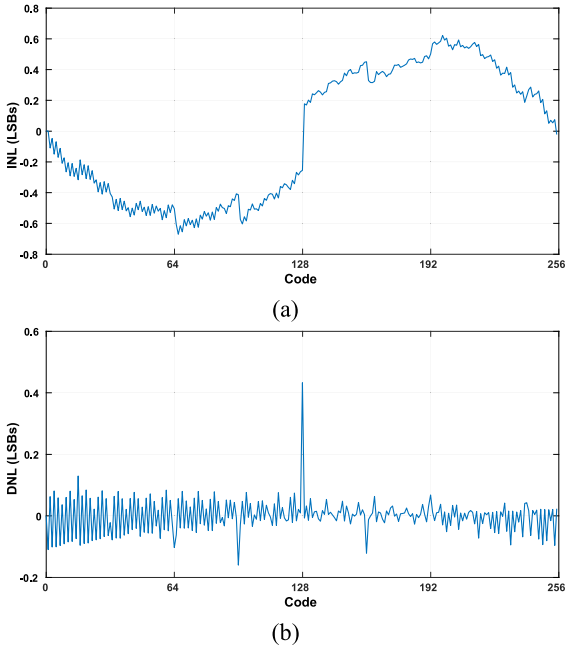


Fig. 17. Measured DAC: (a) INL and (b) DNL.

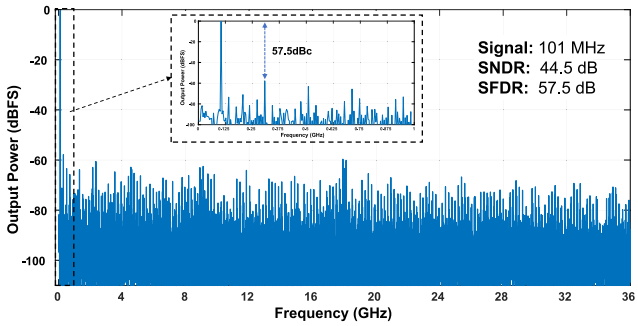


Fig. 18. Measured DAC output spectrum for a 101-MHz sinusoidal tone.

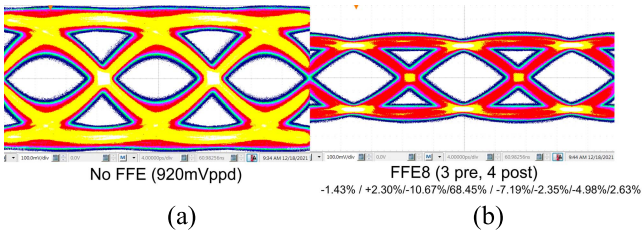


Fig. 19. Measured 72-Gb/s NRZ eye diagrams: (a) without FFE and (b) with eight-tap FFE ($x = 4$ ps/div, $y = 100$ mV/div).

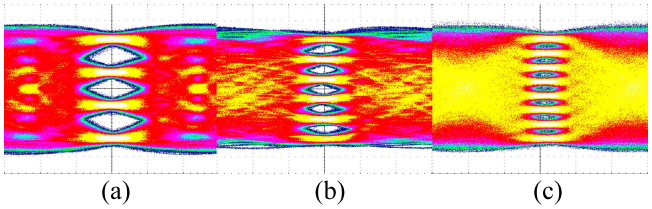


Fig. 20. Measured 72-GBaud PAM eye diagrams. (a) 144-Gb/s PAM4 QPRBS13. (b) 180-Gb/s PAM6 PRBS11. (c) 216-Gb/s PAM8 PRBS15 ($x = 3$ ps/div, $y = 75$ mV/div).

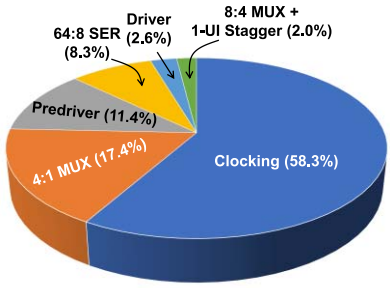


Fig. 21. 72-GBaud TX power breakdown estimated from *RC*-extracted simulations.

B. DAC Characterization

Fig. 17 reports DAC calibrated static linearity, showing an integral nonlinearity (INL) of between −0.63 and +0.67 LSBs, and a differential nonlinearity (DNL) of between −0.16 and +0.43 LSBs. Calibration was performed through external dc measurements of the TX output resistance to determine proper calibration settings for the DAC A and B segments. The calibrated linearity for the 8-b DAC demonstrates the effectiveness of the hybrid digital and analog tuning method for adjusting the weights of the SST driver B segments relative to the SST driver A segments. The DAC output spectrum when producing a 101-MHz sinusoidal output is depicted in Fig. 18. The spectrum was obtained by taking an FFT of output samples captured by a Tektronix DSA8300 sub-sampling oscilloscope and suggests an spurious free dynamic range (SFDR) of 57.5 dB and signal to noise and distortion ratio (SNDR) of 44.5 dB. High-frequency SNDR was limited by an estimated total jitter of $633\text{fs}_{\text{rms}}$. Jitter decomposition analysis shows an estimated $542\text{fs}_{\text{rms}}$ supply-induced jitter, with ±1.15-ps peaks synchronized to internal C64 and C32 sub-rate clock frequencies. Measured SNDR values of 34.9 and 27.3 dB for output sinusoidal tones of 4.46 and 13.46 GHz, respectively, are consistent with this level of phase modulation. The supply noise is attributed to clock buffer switching activity in the 64:8 serializer, where a large clock fan-out is required to capture an aggregate 576 Gb/s from the pattern generator. It is noted that the entire test chip operates from a single 950-mV power supply without any supply regulation for C4 clock buffers.

C. Time-Domain Modulation Experiments

72-Gb/s NRZ eye diagrams with a PRBS15 pattern are seen without FFE in Fig. 19(a) and with eight-tap FFE in Fig. 19(b). Without FFE, a maximum output swing of 920 mVppd is achieved. Fig. 20 shows 72-GS/s TX eye diagrams for 144-Gb/s PAM4 QPRBS13, 180-Gb/s PAM6 PRBS11, and 216-Gb/s PAM8 PRBS15. In all cases, pre-equalized data are loaded into the 32-kB SRAM with eight-tap FFE coefficients listed at the bottom of Fig. 19(b). The relative level mismatch (RLM) for PAM4, PAM6, and PAM8 data is measured at 0.988, 0.981, and 0.978, respectively. The measured power consumption is 288 mW from a 950-mV power supply, with an estimated power breakdown based on an *RC*-extracted simulation shown in Fig. 21.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUMMARY AND COMPARISON AGAINST STATE OF THE ART

	[2]	[4]	[5]	[7]	[15]	[6]	This Work		
Technology Node	14nm	40nm	28nm	7nm	7nm	10nm	4nm		
Symbol Rate (GBaud)	64	100	100	56	66 (97GS/s)	112	72		
Architecture, MUX, & Driver Topology	Analog FFE3, 4:1, CML	7b DAC, 2:1, CML	Analog FFE5, 4:1, CML	8b DAC, 4:1, SST	8b DAC, 4:1, CML	7b DAC, 4:1, CML	8b DAC, 4:1, SST		
Maximum Output Swing (V_{ppd})	1.0	0.56	0.8	0.92	1.1	1.0	0.92		
Active Area (mm^2)	0.048	0.504	0.432	0.032	0.5875*	0.088	0.047		
ESD	Yes	No	No	Yes	Yes	Yes	Yes		
Modulation	PAM4	NRZ	PAM4	PAM4	PAM4	PAM4	PAM4	PAM8	OFDM 64QAM
RLM	0.986	N/A	0.99	0.965	0.996	0.99	0.988	0.978	N/A
Data Rate (Gb/s)	128	100	200	112	132	224	144	216	212
Efficiency (pJ/bit)	1.3	6.19	4.63	1.4	1.94**	1.74 (w/o PLL)	2.0	1.33	1.36

* Single lane + amortized PLL ** Estimated from DAC FoM, SNDR, and 3dB BW, not including PLL power

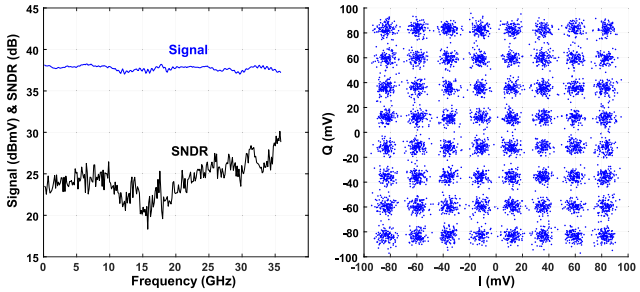


Fig. 22. OFDM experiment with constant power across subchannels. (a) Signal power and SNDR. (b) Estimated QAM64 constellation.

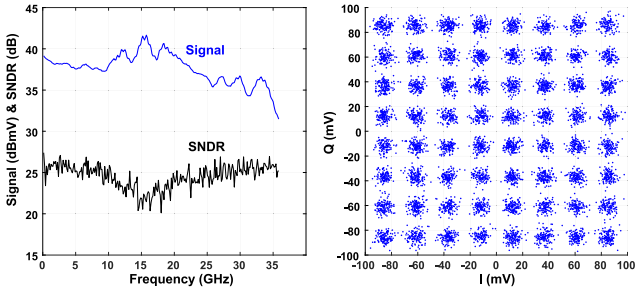


Fig. 23. OFDM experiment with increased signal power near the midband. (a) Signal power and SNDR. (b) Estimated QAM64 constellation.

D. Frequency-Domain Modulation Experiments

To assess TX performance in OFDM applications, QAM64 data were loaded onto 255 OFDM subcarriers from 140 MHz to 35.9 GHz for an aggregate data rate of 212 Gb/s. The OFDM frames employ block-coded modulation (BCM) with an inner binary product code and outer Reed-Solomon KR4 code, resulting in a payload of 184 Gb/s. TX output symbols were captured at 32 points per symbol using the Tektronix sub-sampling scope. The captured data were then post-processed using a software baseband to recover channel symbols and statistical analysis to estimate subchannel SNDR.

In the first OFDM experiment, constant signal power was applied to each of the 255 subchannels. Fig. 22(a) shows the output signal power and SNDR across all subchannels, while Fig. 22(b) presents the estimated QAM64 constellation. The statistical analysis estimates a raw BER of 0.93E-3 and an average subchannel SNDR of 23.9 dB. This performance is sufficient to decode the block code modulated data with no bit errors. The average SNDR is 1.4 dB better than the SNDR obtained for a 72-GS/s PAM8 with a block code modulation, demonstrating the effectiveness of OFDM to overcome jitter limitations compared to time-domain modulation.

The data in Fig. 22 show an SNDR degradation near the midband, which led to a second OFDM experiment with increased signal power in this frequency range, as illustrated in Fig. 23(a). The resulting SNDR showed a modest improvement of 0.4 dB, up to 24.3 dB compared to the first experiment with constant signal power across frequency. The estimated QAM64 constellation is seen in Fig. 23(a), with a raw BER of 0.42E-3. SNDR degradation is still observed in the midband and is attributed to signal distortion in this frequency range.

V. CONCLUSION

A high-speed serial TX in 4-nm FinFET CMOS technology has been reported. Table I summarizes the design's

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.

performance and compares it against that of state-of-the-art high-speed serial TXs [2], [4], [5], [6], [7], [15]. To the best of our knowledge, this is the fastest reported CMOS serial TX employing a voltage-mode driver topology and is the first TX to demonstrate OFDM at these symbol rates. The work shows promise for future wireline links using frequency-domain modulation. At similar line rates, OFDM requires lower DAC and ADC sampling rates than PAM4 and can achieve higher SNR than PAM8. This work demonstrates that CMOS high-speed DACs can achieve sufficient linearity to support frequency-domain modulation in serial links.

ACKNOWLEDGMENT

The authors would like to acknowledge the Samsung Electronics intellectual property (IP) Development Team. The authors are also grateful to Abul Hassan Savanur, Aruldas Divakaran, and the Samsung Semiconductor India Research (SSIR) Layout Team for their support.

REFERENCES

[1] *Beyond 400 Gb/s Ethernet*. Accessed: Aug. 26, 2022. [Online]. Available: https://www.ieee802.org/3/ad_hoc/ngrates/public/calls/20_1029/CFI_Beyond400GbE_201029.pdf

[2] Z. Toprak-Deniz et al., "A 128 Gb/s 1.3 pJ/b PAM-4 transmitter with reconfigurable 3-tap FFE in 14 nm CMOS," *IEEE J. Solid-State Circuits*, vol. 55, no. 1, pp. 19–26, Jan. 2020.

[3] T. O. Dickson, H. A. Ainspan, and M. Meghelli, "A 1.8 pJ/b 56 Gb/s PAM-4 transmitter with fractionally spaced FFE in 14 nm CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2017, pp. 118–119.

[4] P.-J. Peng, S.-T. Lai, W.-H. Wang, C.-W. Lin, W.-C. Huang, and T. Shih, "A 100 Gb/s NRZ transmitter with 8-tap FFE using a 7b DAC in 40 nm CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 130–131.

[5] M. Choi et al., "An output-bandwidth-optimized 200 Gb/s PAM-4 100 Gb/s NRZ transmitter with 5-tap FFE in 28 nm CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2021, pp. 128–129.

[6] J. Kim et al., "A 224-Gb/s DAC-based PAM-4 quarter-rate transmitter with 8-tap FFE in 10-nm FinFET," *IEEE J. Solid-State Circuits*, vol. 57, no. 1, pp. 6–20, Jan. 2022.

[7] M. A. Kossel et al., "An 8b DAC-based SST TX using metal gate resistors with 1.4 pJ/b efficiency at 112 Gb/s PAM-4 and 8-tap FFE in 7 nm CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2021, pp. 130–131.

[8] T. Dickson et al., "A 72 GS/s, 8-bit DAC-based wireline transmitter in 4 nm FinFET CMOS for 200+Gb/s serial links," in *Proc. IEEE Symp. VLSI Technol. Circuits*, Jun. 2022, pp. 28–29.

[9] J. Armstrong, "OFDM for optical communications," *J. Lightw. Technol.*, vol. 27, no. 3, pp. 189–204, Feb. 1, 2009.

[10] A. S. Yonar et al., "An 8-bit 56 GS/s 64× time-interleaved ADC with bootstrapped sampler and class-AB buffer in 4 nm CMOS," in *Proc. IEEE Symp. VLSI Technol. Circuits*, Jun. 2022, pp. 168–169.

[11] C. Menolfi et al., "A 112 Gb/S 2.6 pJ/b 8-tap FFE PAM-4 SST TX in 14 nm CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2018, pp. 104–105.

[12] S. Rylov et al., "A 25 Gb/s ADC-based serial line receiver in 32 nm CMOS SOI," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Jan./Feb. 2016, pp. 56–57.

[13] S. Galal and B. Razavi, "Broadband ESD protection circuits in CMOS technology," *IEEE J. Solid-State Circuits*, vol. 38, no. 12, pp. 2334–2340, Dec. 2003.

[14] L. R. Carley, "A noise-shaping coder topology for 15+ bit converters," *IEEE J. Solid-State Circuits*, vol. 24, no. 2, pp. 267–273, Apr. 1989.

[15] R. L. Nguyen et al., "A highly reconfigurable 40–97 GS/s DAC and ADC with 40 GHz AFE bandwidth and sub-35 fJ/conv-step for 400 Gb/s coherent optical applications in 7nm FinFET," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2021, pp. 136–137.



Timothy O. Dickson (Senior Member, IEEE) received the dual B.Sc. degree (Hons.) in electrical and computer engineering and the M.Eng. degree in electrical engineering from the University of Florida, Gainesville, FL, USA, in 1999 and 2002, respectively, and the Ph.D. degree from the University of Toronto, Toronto, ON, Canada, in 2006. His Ph.D. work was in the area of serial transceivers operating up to 80 Gb/s in SiGe BiCMOS technologies.

In 2006, he joined the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA, where he is currently a Research Scientist. He is also an Adjunct Professor with Columbia University, New York, NY, USA, where he has been teaching graduate-level courses in analog and mixed-signal integrated circuit design since 2007. His research focuses on the design of high-speed, low-power serial transceivers for electrical and optical links.

Dr. Dickson was a member of the Technical Program Committee (TPC) of the IEEE Compound Semiconductor Integrated Circuit Symposium from 2007 to 2009. He was a recipient or co-recipient of several best paper awards, including the Best Paper Award for the 2009 IEEE JOURNAL OF SOLID-STATE CIRCUITS, the Beatrice Winner Award for Editorial Excellence at the 2009 IEEE International Solid-State Circuits Conference (ISSCC), the Best Paper Award at the 2015 IEEE Custom Integrated Circuits Conference (CICC), and the Best Student Paper Award at the 2004 Symposium on VLSI Circuits Since 2017, he has been serving on the CICC TPC, where he previously chaired the Wireline Subcommittee and is a CICC Fellow. He was a Guest Editor of the October 2010 Issue of the IEEE JOURNAL OF SOLID-STATE CIRCUITS. Since 2018, he has been an Associate Editor for the IEEE SOLID-STATE CIRCUITS LETTERS. From 2019 to 2020, he has served as a Distinguished Lecturer for the IEEE Solid-State Circuits Society.



Zeynep Toprak Deniz (Senior Member, IEEE) received the B.S. degree from Istanbul Technical University, Istanbul, Turkey, in 1999, the M.S. degree from SabancıUniversity, Istanbul, in 2001, and the Ph.D. degree from the Swiss Federal Institute of Technology (EPFL), Lausanne, Switzerland, in 2006, all in electrical engineering.

From 1999 to 2003, she worked at a design center in Istanbul, Turkey, on various projects, including dc–dc converters, VCXOs, and low-jitter phase-locked loop (PLL)-based clock and data recovery (CDR) circuits. In January 2003, she joined EPFL as a Graduate Student, working at the Microelectronic Systems Laboratory as a Research Assistant. In her doctoral work, she designed and demonstrated a unique analog computation system for global energy management and optimization in Network-on-Chip applications in her doctoral work. In 2007, she joined the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA, as a Research Staff Member. She has published more than 30 papers in IEEE journals or conferences. She holds 15 issued U.S. patents. Her research interests include the design of mixed-signal CMOS circuits for high-speed data communications, power-efficient and high-speed analog-to-digital converters, and voltage regulators.



Martin Cochet (Senior Member, IEEE) received the Dipl.Ing. degree from the École Polytechnique, Palaiseau, France, in 2012, the M.S. degree from Imperial College London, London, U.K., in 2013, and the Ph.D. degree in electronic engineering from Aix-Marseille University, Marseille, France, in 2016.

In 2015, he was a Visiting Scholar with the University of California at Berkeley, Berkeley, CA, USA. From 2013 to 2016, he was with STMicroelectronics, Crolles, France. He is currently a Research Scientist with the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA. He holds five U.S. patents. His research interests include mixed-signal circuit design for high-speed wireline communications and SoC power management.

Dr. Cochet received the 2016 A-SSCC Best Student Demonstration and the Aix-Marseille University Distinguished Ph.D. Dissertation Award. He serves on the Technical Program Committee of the European Solid-State Circuit Conference.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.



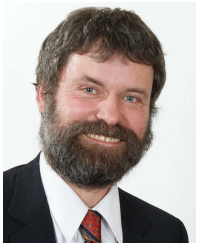
Troy J. Beukema received the B.S.E.E. and M.S.E.E. degrees from Michigan Technological University, Houghton, MI, USA, in 1984 and 1988, respectively.

Since 1996, he has been with IBM Research, Yorktown Heights, NY, USA, where he has focused on communication system design and analysis for WLAN, 60-GHz wireless systems, and high-speed wireline CMOS serializer–deserializer (SerDes). His work in wireline systems spans data rates from 6 to 100 Gb/s and beyond for applications in IBM systems and technology partner electrical transceivers. His ongoing research activity is in the area of advanced modulation, equalization, and coding for future 200-Gb/s wireline systems.



Marcel Kossel (Senior Member, IEEE) received the Dipl.Ing. and Ph.D. degrees in electrical engineering from the Swiss Federal Institute of Technology (ETH), Zürich, Switzerland, in 1997 and 2000, respectively.

He joined IBM Research Zurich, Rüschlikon, Switzerland, in 2001, where he is involved in analog circuit design for high-speed serial links. He has also worked in the field of microwave tagging systems and radio frequency identification systems. His research interests include analog circuit design and RF measurement techniques.



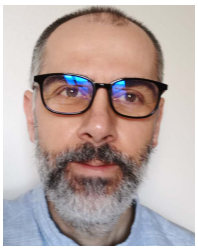
Thomas Morf (Senior Member, IEEE) received the B.S. degree from the Zürich University of Applied Science, Zürich, Switzerland, in 1987, the M.S. degree in electrical and computer engineering from the University of California at Santa Barbara (UCSB), Santa Barbara, CA, USA, in 1991, and the Ph.D. degree from the Swiss Federal Institute of Technology (ETH), Zürich, in 1996.

From 1989 to 1991, he worked as a Research Assistant at UCSB, performing research in the field of active microwave inductors and digital GaAs circuits. In 1991, he joined ETH, where, in his doctoral work, he investigated circuit design and processing for high-speed optical links on GaAs using epitaxial liftoff techniques. In 1996, he transferred to the Electronics Laboratory, ETH, where he led a research group in the area of InP HBT analog circuit design and process technology. In 1999, he joined IBM Research Zurich, Rüschlikon, Switzerland. He has coauthored more than 190 papers and is a co-inventor of more than 45 issued patents. His current research interests include electrostatic discharge (ESD) circuit protection, electrical and optical high-speed high-density interconnects, on-chip coil modeling, and cryogenic electronics for quantum computing.



Young-Ho Choi received the bachelor's degree in electrical engineering and computer science from Kyungpook National University, Daegu, South Korea, in 2008, and the master's and Ph.D. degrees in electrical engineering from the Pohang University of Science and Technology (POSTECH), Pohang, South Korea, in 2016.

He joined Samsung Electronics, Hwaseong, South Korea, in 2016, where he has been involved in the design of high-speed wireline serializer–deserializer (SerDes), such as gigabit Ethernet and PCI Express. His interests include high-speed chip-to-chip interface circuits, clock and data recovery (CDR), clocking circuits, signal/power integrity, and next-generation SerDes architecture.



Pier Andrea Francese (Senior Member, IEEE) received the Laurea degree in electrical engineering (cum laude) from the Politecnico di Milano, Milan, Italy, in 1993, and the Ph.D. degree from the Federal Institute of Technology of Zürich (ETH), Zürich, Switzerland, in 2005.

He is currently a Principal Research Staff Member with the IBM Zürich Research Laboratory, Rüschlikon, Switzerland, where he is the Technical Leader of the High-Speed Interconnect Technology Group. His research interests are in the areas of high-speed data converters, analog equalization, and clock-data-recovery circuit techniques in advanced CMOS technology nodes.



Matthias Brändli received the Dipl.Ing. (M.Sc.) degree in electrical engineering from the Swiss Federal Institute of Technology (ETH), Zürich, Switzerland, in 1997.

From 1998 to 2001, he was with the Integrated Systems Laboratory, Swiss Federal Institute of Technology (ETH), working on deep-submicrometer technology VLSI design challenges, digital video image processing for biomedical applications, and testability of CMOS circuits. In 2001, he joined the Microelectronics Design Center, ETH Zürich, Zürich, where he was involved in numerous digital and mixed-signal ASIC design projects, worked on EDA design automation, and contributed to teaching. In 2008, he joined the IBM Zürich Research Laboratory, Rüschlikon, Switzerland, where he has been working on multi-gigabit/s, low-power communication circuits in advanced CMOS technologies.



Christian W. Baks received the B.S. degree in applied physics from the Fontys College of Technology, Eindhoven, The Netherlands, in 2000, and the M.S. degree in physics from the State University of New York, Albany, NY, USA, in 2001.

In 2001, he joined the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA, as an Engineer, where he is involved in high-speed optoelectronic package and backplane interconnect design specializing in signal integrity issues.



Jonathan E. Proesel (Senior Member, IEEE) received the B.S. degree in computer engineering from the University of Illinois at Urbana–Champaign, Urbana, IL, USA, in 2004, and the M.S. and Ph.D. degrees in electrical and computer engineering from Carnegie Mellon University, Pittsburgh, PA, USA, in 2008 and 2010, respectively.

From 2010 to 2020, he was a Research Staff Member in communications circuits and systems with the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA. He held internships with IBM Microelectronics, Essex Junction, VT, USA, in 2004, and IBM Research, Yorktown Heights, in 2009. He is currently the IC Team Lead and a Principal Engineer with Nubis Communications, New Providence, NJ, USA, where he works on analog and mixed-signal circuit design for optical transmitters and receivers. His research interests include high-speed optical and electrical communications, silicon photonics, data converters, and bioelectronics.

Dr. Proesel is a Senior Member of the IEEE Solid-State Circuits Society. He was a recipient of the Analog Devices Outstanding Student Designer Award in 2008 and the SRC Techcon Best in Session Award for Analog Circuits in 2009, and a co-recipient of the Best Student Paper Award at the 2010 IEEE Custom Integrated Circuits Conference and the 2018 IEEE PHOTONICS TECHNOLOGY LETTERS Best Paper Award. He received multiple IBM technical awards. He has served on the Technical Program Committee of the Symposium on VLSI Circuits from 2018 to 2021.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.



John F. Bulzacchelli (Member, IEEE) was born in New York, NY, USA, in 1966. He received the B.S., M.S., and Ph.D. degrees in electrical engineering from the Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, in 1990, 1990, and 2003, respectively.

From 1988 to 1990, he was a Co-Op Student with Analog Devices, Wilmington, MA, USA, where he invented a new type of delay-and-phase-locked loop for high-speed clock recovery. From 1992 to 2002, he conducted his doctoral research at the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA, in a joint study program between IBM and MIT. In his doctoral work, he designed and demonstrated a superconducting bandpass delta–sigma modulator for direct ADC conversion of multi-GHz RF signals. In 2003, he became a Research Staff Member at the IBM Thomas J. Watson Research Center, where he focused on the design of mixed-signal CMOS circuits for high-speed data communications. In the last few years, he has resumed research activities in cryogenic electronics, investigating the use of both cryo-CMOS and SFQ circuits for quantum computing control and readout. He holds 42 U.S. patents.

Dr. Bulzacchelli received the Jack Kilby Award for Outstanding Student Paper at the 2002 IEEE International Solid-State Circuits Conference (ISSCC), was a co-recipient of the Beatrice Winner Award for Editorial Excellence at the 2009 ISSCC, coauthored the IEEE JOURNAL OF SOLID-STATE CIRCUITS article awarded the Best Paper of 2009, and was a co-recipient of the IEEE Custom Integrated Circuits Conference (CICC) Best Regular Paper Awards in 2011 and 2015.



Michael P. Beakes (Member, IEEE) received the B.A. degree in physics from SUNY Potsdam, Potsdam, NY, USA, the B.S. degree in electrical engineering from Clarkson University, Potsdam, NY, USA, in 1980, and the M.S. degree in computer engineering from Syracuse University, Syracuse, NY, USA, in 1984.

He retired from his position as a Senior Scientist at the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA, in 2021. Throughout his 41-year career at IBM, he concurrently played the roles of a Designer, a Methodology/Tools Developer, and a Team Leader for the RF/Analog/Mixed-Signal Integrated Circuit (IC) Design Community. He is also a mentor and an advisor for grade school and high school technology and robotics programs.



Byoung-Joo Yoo (Member, IEEE) received the B.S. degree in electrical engineering from Korea University, Seoul, South Korea, in 2005, and the M.S. and Ph.D. degrees in electrical engineering from Seoul National University, Seoul, in 2007 and 2013, respectively.

He is currently a Principle Engineer with Samsung Electronics, Hwaseong, South Korea. His current research focuses on the high-speed serializer–deserializer (SerDes) architecture for next-generation wireline links. His expertise mainly includes digital and analog integrated circuit design for die-to-die interfaces and giga-bit Ethernet protocols.



Hyoungbae Ahn (Member, IEEE) received the B.Sc. and Ph.D. degrees in electrical and computer engineering from Seoul National University, Seoul, South Korea, in 2014 and 2020, respectively.

He currently works at Samsung Electronics, Hwaseong, South Korea, designing high-speed serializer–deserializer (SerDes) DSP.



Dong-Hyuk Lim received the B.S., M.S., and Ph.D. degrees in electrical engineering and computer science from Seoul National University, Seoul, South Korea, in 2006, 2008, and 2013, respectively.

Since 2013, he has been with Samsung Electronics Corporation, Hwaseong, South Korea, developing high-speed wireline transceivers. He is currently a Principal Engineer and leads a PHY analog group at the Samsung Foundry intellectual property (IP) Development Team, Hwasong, South Korea. He has contributed to serializer–deserializer (SerDes) architecture, system modeling, link simulation, and developing SerDes APIs for DSP-based four-level pulse amplitude modulation (PAM4) transceivers.



Gunil Kang joined Samsung Electronics, Hwaseong, South Korea, in 2006. He is currently a technical lead, leading a digital design and DSP architecture on the research and development of four-level pulse amplitude modulation (PAM4) DSP ASICs. He has more than 16 years of experience in digital communications, signal processing, and digital design for serial links.



Sang-Hune Park received the B.S., M.S., and Ph.D. degrees in electronics and electrical engineering from the Pohang University of Science and Technology, Pohang, South Korea, in 2000, 2002, and 2008, respectively.

He joined Samsung Electronics, Hwaseong, South Korea, in 2008, where he is currently the Vice-President of the Foundry Division. His research interests include high-speed DRAM interface, low-power serializer–deserializer (SerDes), hetero integration chiplet technology, and intellectual property (IP) ecosystem.



Mounir Meghelli (Member, IEEE) received the M.S. degree in electronics and automatics from the University of Paris, Orsay, France, in 1992, the Engineering degree in telecommunication from ENST, Paris, France, in 1994, and the Ph.D. degree from the University of Paris VI, Paris, after a four-year research program with the CNET France Telecom Research Center, Fishkill, NY, working on the design of high-speed ICs for optical communications in GaAs and InP HBT technologies.

From 1998 to 2005, he was a Research Staff Member with the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA, working on the design of high-frequency ICs in SiGe BiCMOS and CMOS technologies for wireline and wireless applications. From 2006 to 2011, he was with the IBM Server and Technology Group where he held the position of Senior Technical Staff Member position leading the design of advanced serial links for storage, networking, and server applications. Since 2012, he has been managing the Mixed Signal Communication IC Design Group, IBM Thomas J. Watson Research Center.



Hyo Gyuem Rhew received the B.S. degree (summa cum laude) from Seoul National University, Seoul, South Korea, in 2006, and the M.S. and Ph.D. degrees from the University of Michigan, Ann Arbor, MI, USA, in 2009 and 2012, respectively, all in electrical engineering.

From 2012 to 2020, he was with Broadcom, Irvine, CA, USA, where he built various types of high-speed wireline transceivers. In 2020, he joined Samsung Electronics, Hwaseong, South Korea, and currently leads a serializer–deserializer (SerDes)/PHY group at the Foundry Business Division as the Vice-President. His research interests include SerDes circuits for Ethernet, PCIe, optical, and other applications.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.



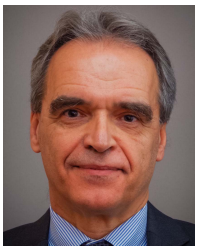
Daniel J. Friedman (Fellow, IEEE) is currently a Distinguished Research Staff Member and a Senior Manager of the Communication Circuits and Systems Department, IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA. At IBM, he initially developed field-powered RFID tags before turning to high data rate wireline and wireless communication. He holds more than 90 patents and has authored or coauthored more than 75 publications. His current research interests include high-speed I/O design, phase-locked loop (PLL) design, mmWave circuits and systems, accelerator designs, and circuit/system approaches to enabling new computing paradigms.

Dr. Friedman was a member of the BCTM Technical Program Committee from 2003 to 2008 and the ISSCC International Technical Program Committee of IEEE International Solid-State Circuits Conference (ISSCC) from 2009 to 2016. He was a member-at-large of the SSCS Adcom from 2018 to 2020. He was a co-recipient of the Beatrice Winner Award for Editorial Excellence at the 2009 ISSCC, the 2009 JSSC Best Paper Award (given in 2011), the 2017 ISSCC Lewis Winner Outstanding Paper Award, and the 2017 JSSC Best Paper Award (given in 2019). He has served as the Wireline Sub-Committee Chair of ISSCC from 2012 to 2016. He has served as the SSCS Distinguished Lecture Chair from 2020 to 2021. Since 2017, he has been serving as the ISSCC Short Course Chair. He was an SSCS Distinguished Lecturer in 2017 and 2018. He has been serving as an Associate Editor for the JSSC since mid-2019. Since 2021, he has been serving as the SSCS Vice President of Membership.



Michael Choi received the master's and Ph.D. degrees from the University of California at Los Angeles, Los Angeles, CA, USA, in 1998 and 2002, respectively.

He joined Samsung Electronics, Hwaseong, South Korea, in 2006, where he is a Master (VP of Technology). His expertise includes high-speed data converters and various analog front ends for UHD digital TV, Wi-Fi & 5G connectivity, automotive V2X, touch controller, and CMOS image sensor.



Mehmet Soyuer (Life Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of California at Berkeley, Berkeley, CA, USA, in 1988.

He then joined the IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA, as a Research Staff Member. His work has involved high-frequency mixed-signal integrated circuit designs, in particular, monolithic phase locked-loop designs for clock and data recovery, clock multiplication, and frequency synthesis using silicon and SiGe technologies. At the IBM Thomas J. Watson Research Center, he managed the Mixed-Signal Communications Integrated-Circuit Design Group from 1997 to 2000. He was the Senior Manager of the Communication Circuits and Systems Department, IBM Thomas J. Watson Research Center, from 2000 to 2006. In 2006, he has been promoted to the position of Department Group Manager of Communication and Computation Subsystems at the Thomas J. Watson Research Center. He also holds the title of Distinguished Research Staff Member. He has authored numerous papers in the areas of analog, mixed-signal, RF, microwave, and nonlinear electronic circuit design.

Dr. Soyuer was a Technical Program Committee Member of the Topical Meeting on Silicon Monolithic Integrated Circuits in RF Systems (SiRF) in 2004 and 2006. He has served as an elected member of the Solid-State Circuit Society (SSCS) Administrative Committee (AdCom) for two terms. From 1997 to 2008, he was a Technical Program Committee Member of the International Solid-State Circuits Conference (ISSCC). He was an Associate Editor of the IEEE JOURNAL OF SOLID-STATE CIRCUITS from 1998 to 2000 and was one of the guest editors for the December 2003 Special ISSCC Issue. He chaired the Analog, MEMS and Mixed-Signal Electronics Committee of the International Symposium on Low Power Electronics and Design (ISLPED) in 2001. He had served as a Distinguished Lecturer of IEEE SSCS.



Jongshin Shin (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronics and electrical engineering from Seoul National University, Seoul, South Korea, in 1997, 1999, and 2004, respectively.

He joined Samsung Electronics, Hwaseong, South Korea, in 2004, as a member of the Technical Staff, where he is currently the Vice-President of the Foundry Division. His research interests include clock generators, high-speed I/O, and clock and data recovery circuits.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:32:04 UTC from IEEE Xplore. Restrictions apply.